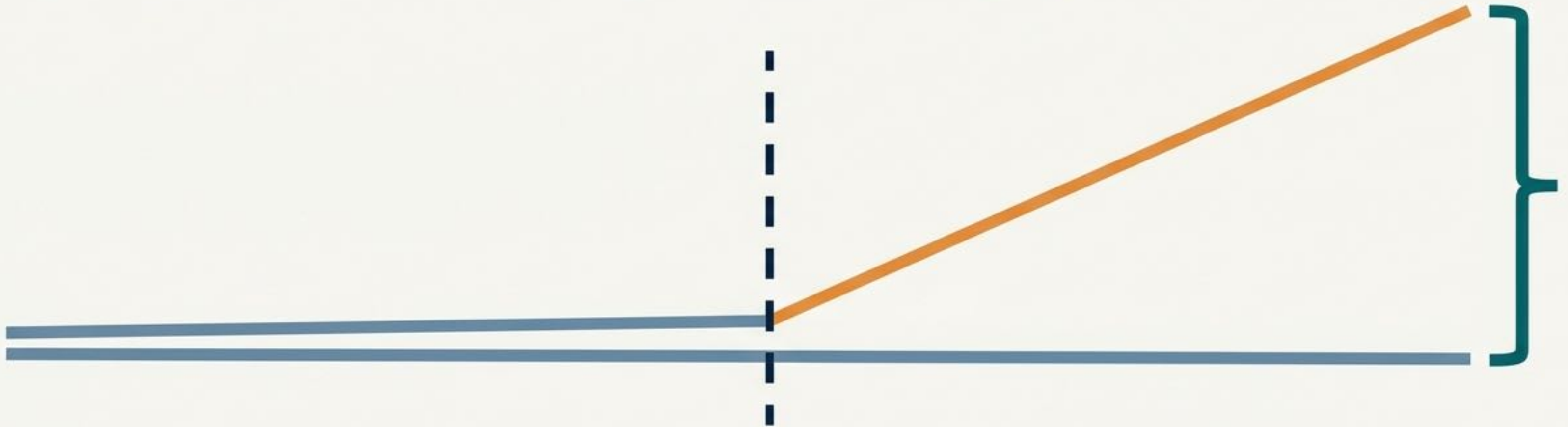


Applied Difference-in-Differences in Python

A Methodological Study Guide for Theory,
Implementation, and Interpretation using PyFixest



Confounding Secular Trends Overstate Naive Estimates

Core Context: An after-school tutoring program is implemented in 10 out of 35 high schools. We measure average GPA for low-income students.

(The Flawed View)

+36.20 Points

Treated schools jumped
from 60.17 to 96.37.

(The Structural Reality)

**+25.32
Points**
(Causal Effect)

**+10.88
Points**
(Secular Trend)

Untreated schools
drifted from
71.22 to 82.10.

Analogy: Blaming the rooster for the sunrise. You need a rooster-free village to separate the rooster's effect from the natural passing of time. DiD nets out the district-wide drift.

Core Econometric Vocabulary for the DiD Framework

ATT

$$E[Y(1) - Y(0) \mid D=1]$$

The causal effect for units that actually got treated.

Intuitive analogy: The extra speed gained by the specific car whose engine was turned on.



Counterfactual

Unobserved

The hypothetical outcome for treated units without treatment (Baseline + Control Drift).

Intuitive analogy: The path the treated track would have taken.



SUTVA

No Interference & Single Version

Stable Unit Treatment Value Assumption. No interference/spillovers between units, and a single version of treatment.

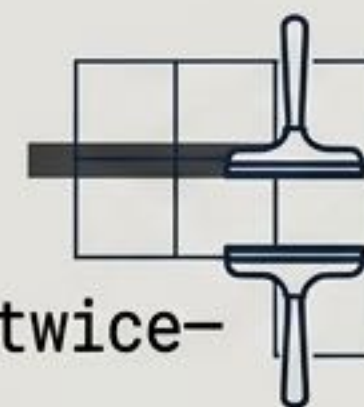


Intuitive: No contagion between patients; sharing medication invalidates the trial.

TWFE

Two-Way Fixed Effects

The regression implementation absorbing unit and time invariances.

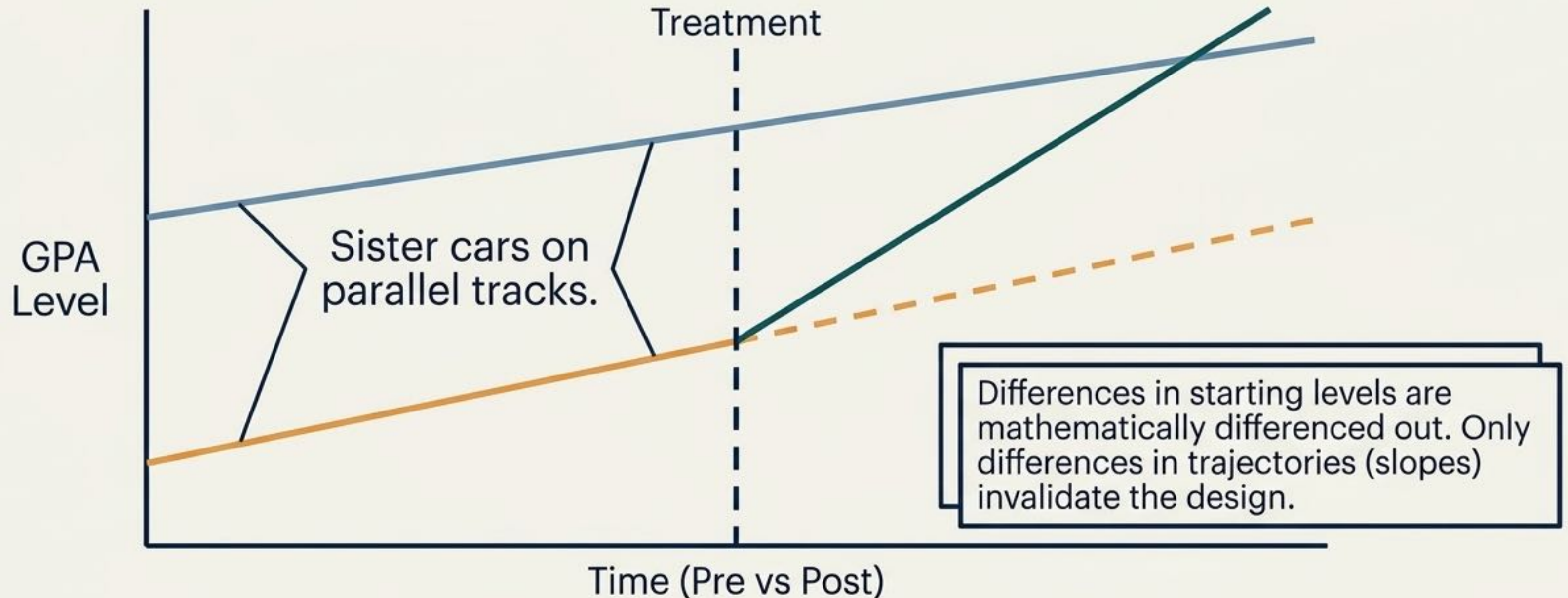


Intuitive: Wiping the negative twice—once for school-specific stains, once for period-wide glare.

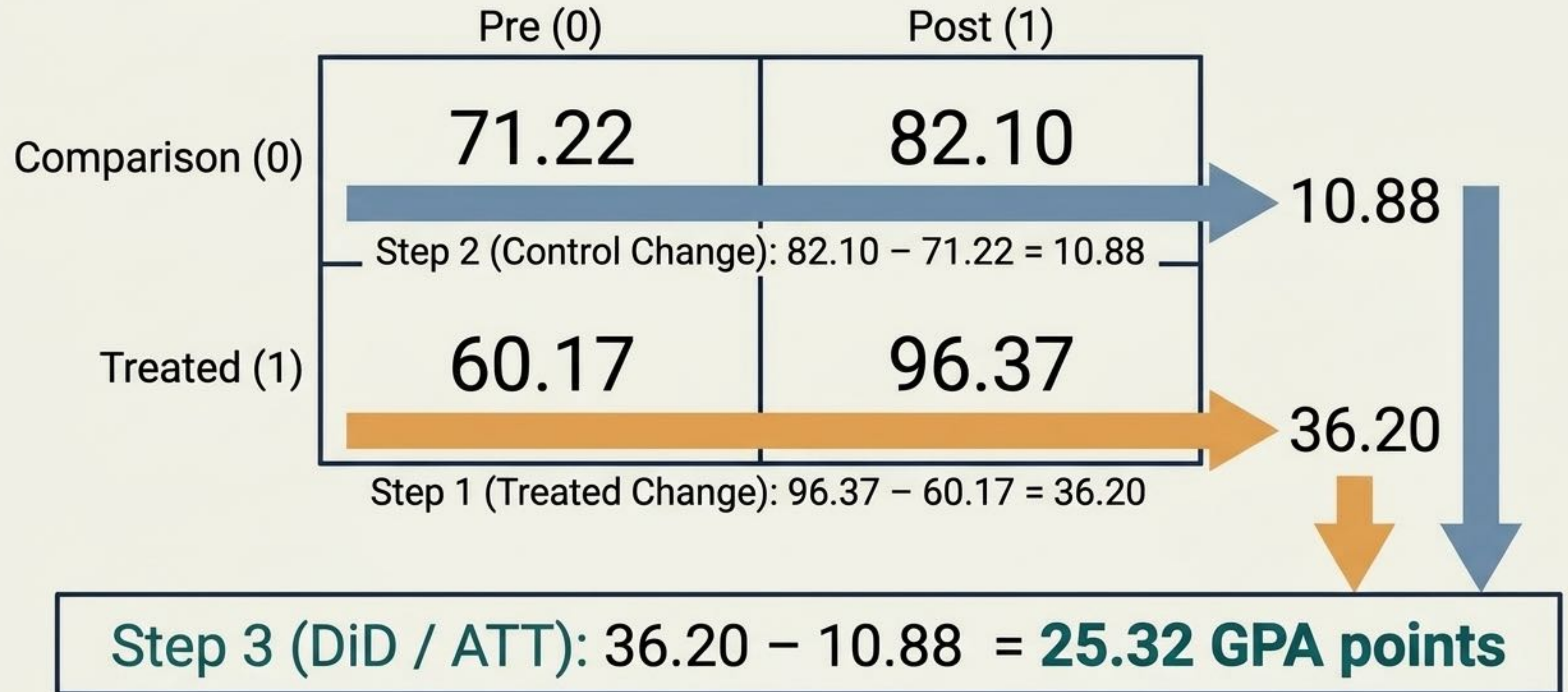
The Identifying Assumption is Parallel Trends

Formal Definition:

$$E[Y_{\{i,1\}}(0) - Y_{\{i,0\}}(0) \mid D=1] = E[Y_{\{i,1\}}(0) - Y_{\{i,0\}}(0) \mid D=0]$$



Mechanics of the 2x2 Estimator



The naive approach overstated the actual 25.32 GPA point effect by 43%.

Translating TWFE Mathematics into Python Syntax

$$Y_{it} = \beta_3(Treat_i \times Post_t) + \gamma_i + \vartheta_t + \epsilon_{it}$$

(Absorbs school-level
starting gaps)

(Absorbs district-wide
time shocks)

```
fit_twfe = pf.feols("gpa ~ txp | id + time", data=df)
```

The coefficient on txp isolates the ATT precisely at **25.315**.

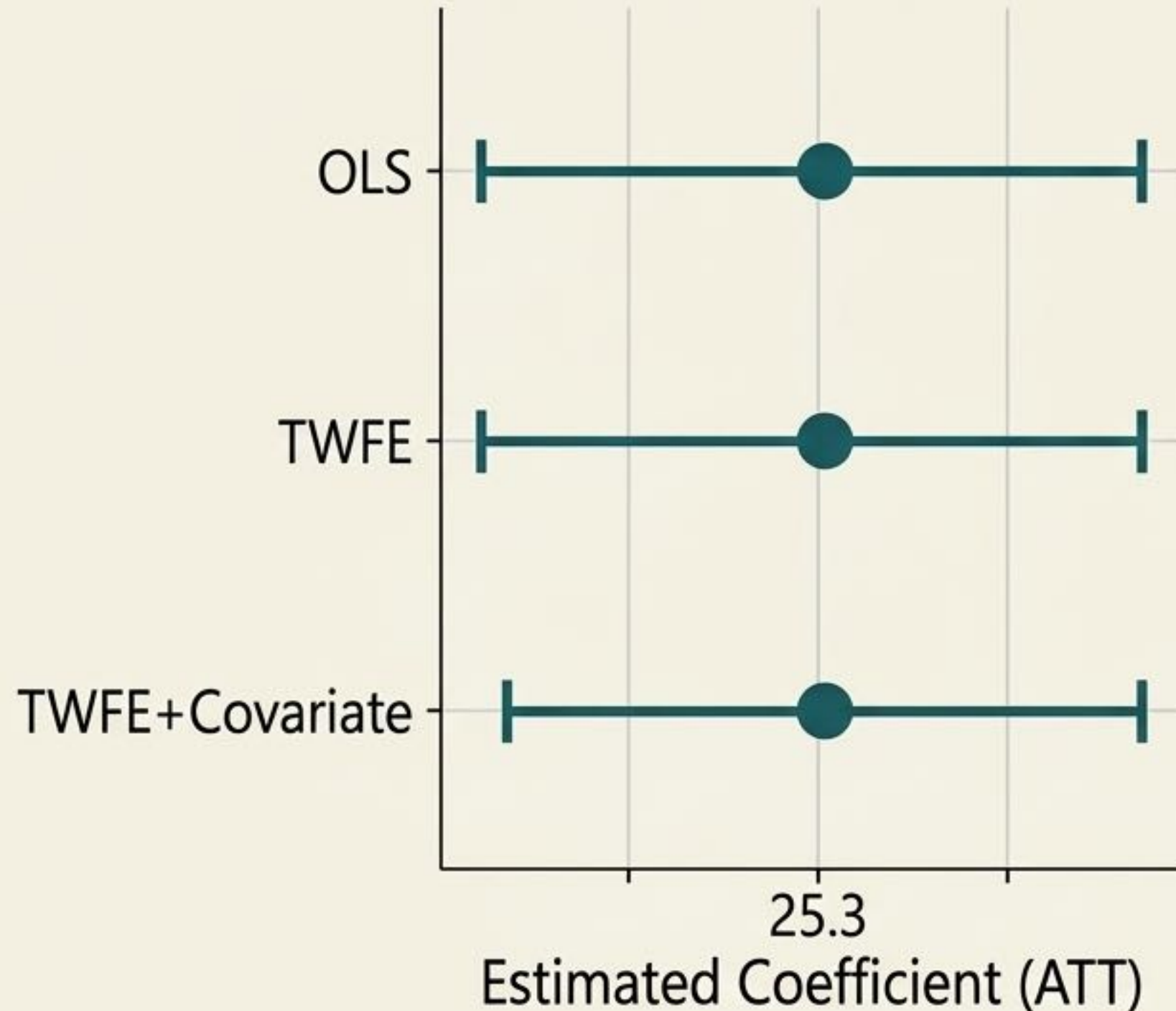
Designing Robust Inference for Panel Data

Estimator	SE	Diagnostic Assumption
iid	0.607	Assumes homoskedasticity (Rarely justified in panel data)
HC1	0.585	White robust; ignores within-cluster correlation
CRV1	0.585	Clustered by school; the workhorse for panel data
CRV3	0.637	Jackknife cluster; corrects CRV1 downward bias

The point estimate remains identical (25.315) across all models. With a massive effect ($t \approx 40$), the choice of SE matters little here.

For smaller effects with limited treated clusters (e.g., 10 schools), CRV3 or wild cluster bootstraps are the safer methodological defaults to prevent false positives.

Testing Robustness with Time-Varying Covariates



```
feols("gpa ~ txp + female_share  
      | id + time")
```

Result: ATT shifts trivially from
25.315 $\rightarrow$ 25.328.

Bad Controls

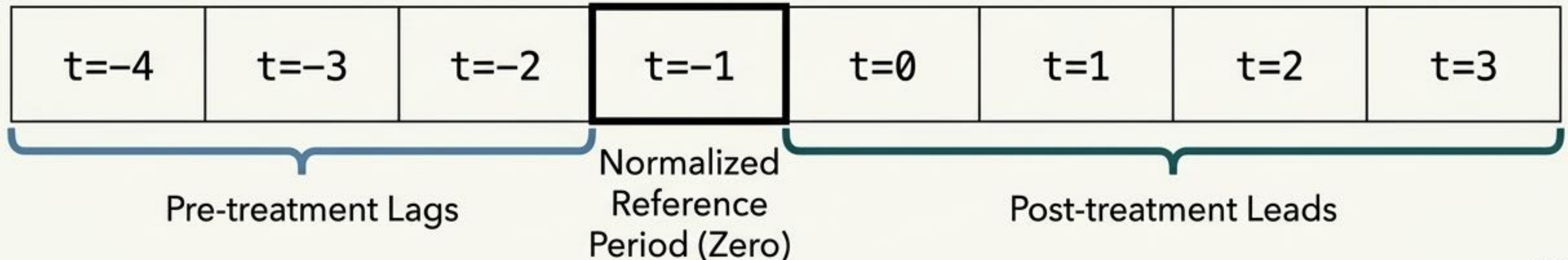
If an intervention alters a covariate (e.g., tutoring attracts a different demographic mid-year), controlling for it absorbs the causal effect. Covariates must be predetermined or strictly exogenous.

Expanding to Dynamic Effects via the Event Study

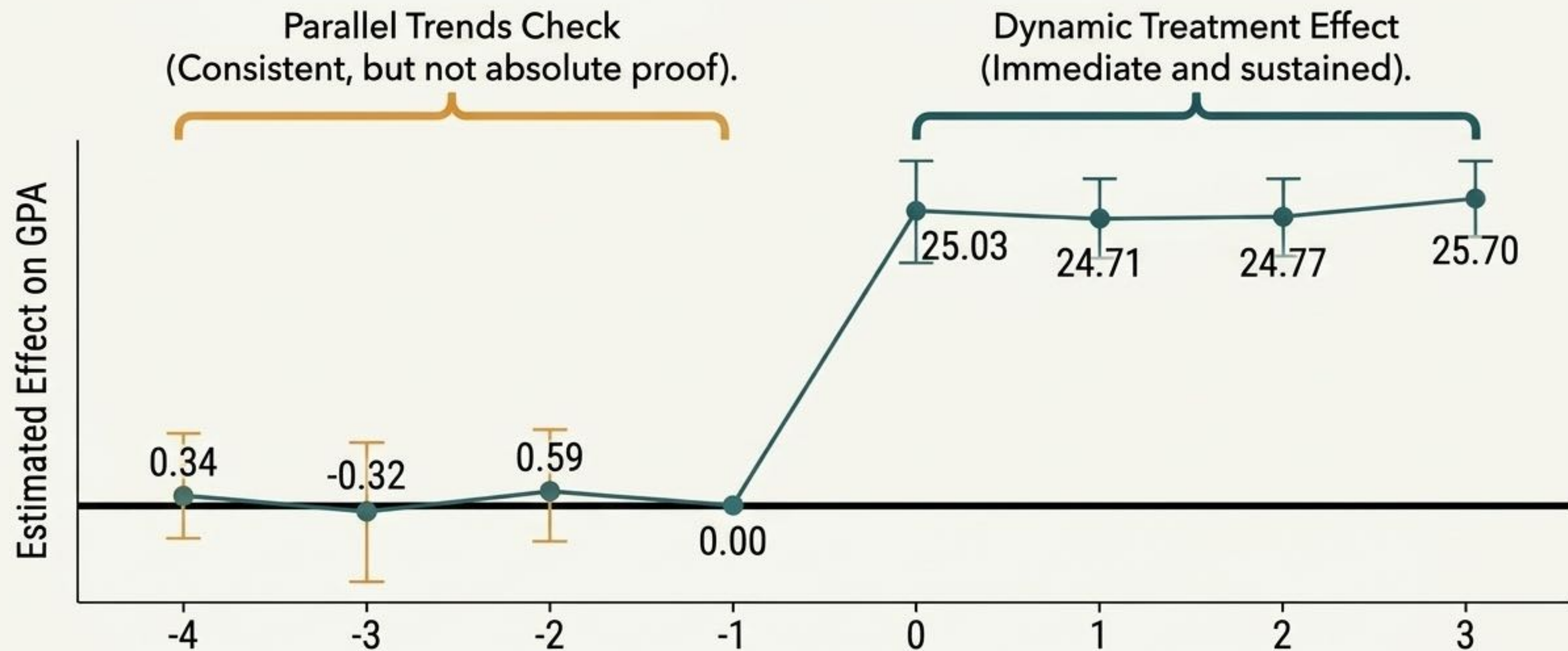
1. Diagnostic: Check pre-treatment periods for parallel trends.
2. Dynamic: Measure if the effect is immediate, delayed, growing, or fading.

```
feols("gpa ~ i(timeToTreat, ref=-1) | id + time")
```

$$Y_{it} = \sum_{j \neq -1} \theta_j \cdot D_i \cdot 1[t - E_i = j] + \dots$$



Interpreting Event Study Dynamics



There is no ramp-up or fade-out. The effect appears fully in the first treated period.

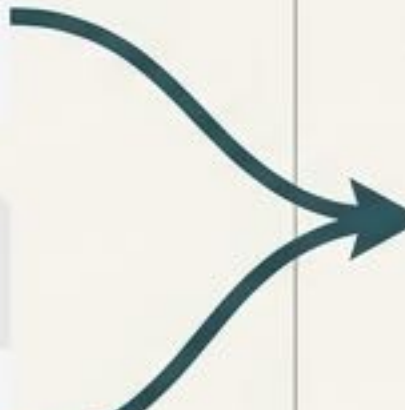
Exporting Publication-Quality Academic Tables

The Stepwise Code

```
csw(txp, female_share)
```

The Export Code

```
pf.etable(models, type="tex",  
  coef_fmt="b* \n(se)")
```



	(1)	(2)
Variables	TWFE	TWFE+Cov
Treatment $\times$ Post	25.315*** (0.585)	25.328*** (0.605)
Female Share		-3.216 (8.700)
Fixed Effects	Yes	Yes

Methodological Caveats and Modern Context



Unrealistic Effect Sizes

An R^2 of 0.995 and a massive 25-point ATT are artifacts of simulated data. Real effects are substantially noisier and smaller.



Staggered Adoption

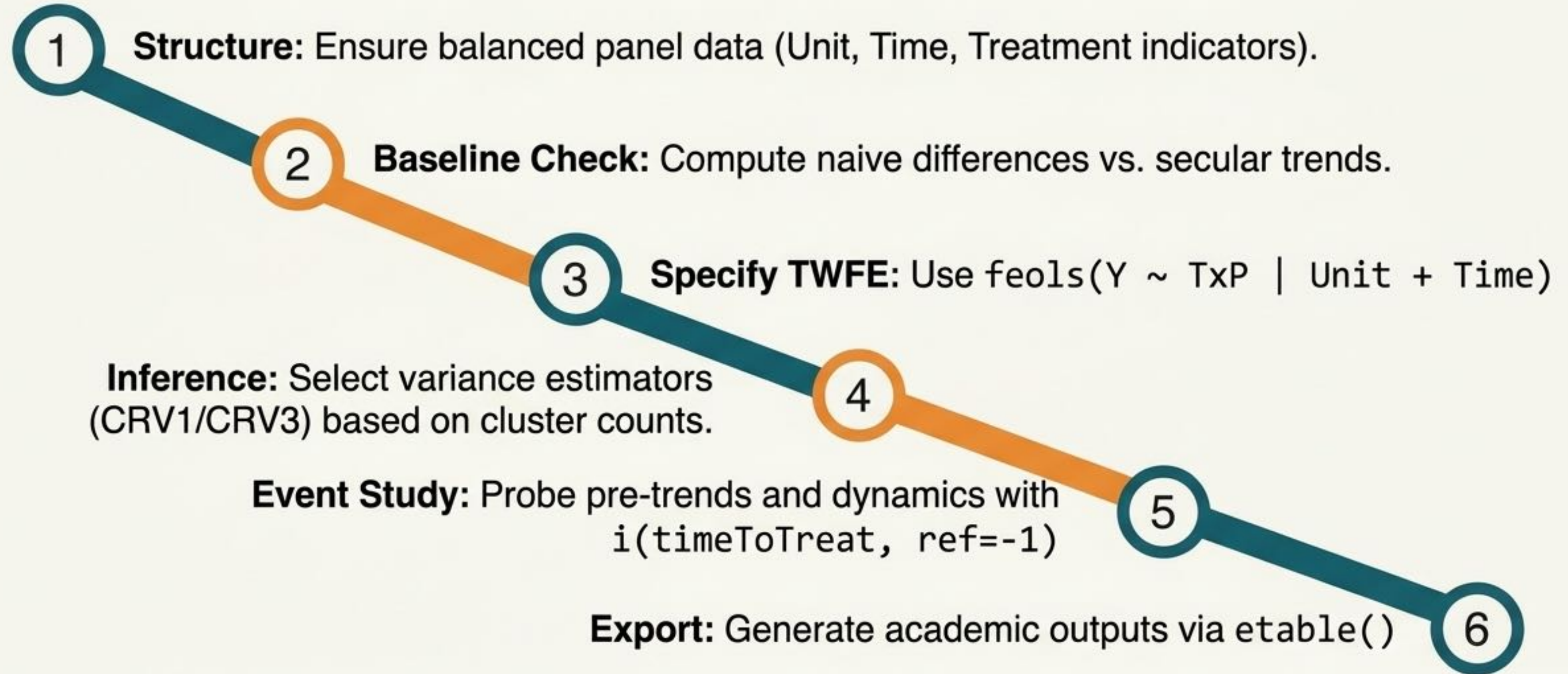
If treatment rolls out at different times with heterogeneous effects, basic TWFE is fundamentally biased. Already-treated units act as “bad controls.” Modern event study solutions are required.



Small Cluster Bias

With only 10 treated clusters, standard CRV1 errors become fragile. Robust inference requires CRV3 or Wild Cluster Bootstraps to avoid false positives.

The Python DiD Implementation Pipeline



A credible control group and clean research design drive valid causal conclusions—code simply measures them